

Week 9 Report: Result Analysis

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

Student: Jayanth Akarapu

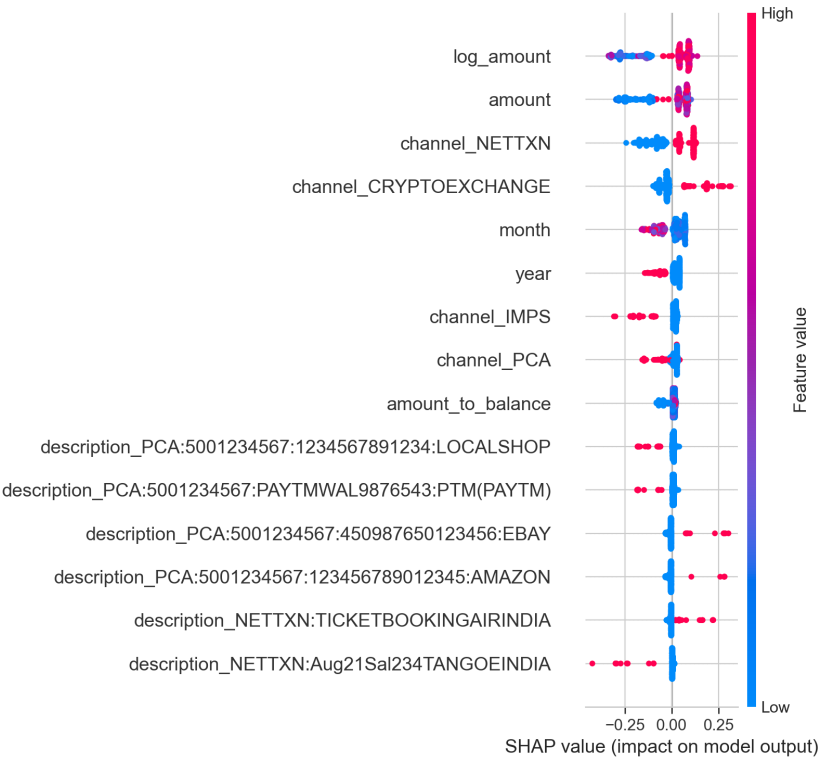
Supervisor: Dr. Prasanth Nayak

What I did this week

This week I analysed how well the results would hold up in practice. When I train on transactions up to 2024-03-16 and test on later ones, the tree models stay near perfect, but SVM drops to F1 0.855 because it leans on date features.

The hardest test holds out whole merchants. On merchants never seen in training, F1 falls to between 0.857 and 0.878. So the near-perfect test scores come mostly from memorising risky merchants. For the explanation I used Random Forest importance and SHAP. The top features are the amount, the log amount and the NETTXN and crypto channels. Training time grows slowly for the trees but quickly for SVM (7.7 s against 0.09 s for Decision Tree on the full training set).

Model	F1 on unseen merchants	ROC-AUC
Decision Tree	0.858	0.963
Random Forest	0.869	0.974
SVM (RBF)	0.857	0.953
Gradient Boosting	0.878	0.974





Training time as the training set grows.

Next week

Start writing the report: introduction, related work and methodology.